$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, $x_i \in \mathbb{R}^n$ ← input data

$y_i \in \mathbb{R}$ ← labels / outputs.

Model:

$y = \alpha_1 f_1(x) + \cdots + \alpha_k f_k(x)$ where $\{f_1, \ldots, f_k\}$ was chosen basis functions

(to explain the unknown relationship $y = f(x)$)

Loss $\mathcal{L}$: $\sum_{i=1}^N (\hat{y}_i - y_i)^2 = \sum_{i=1}^N r_i^2 = \|\Phi\alpha - y\|_2^2$

$\hat{\alpha} \in \mathbb{R}^k$ which minimizes $\mathcal{L}$ satisfies

$$\Phi^T \Phi \hat{\alpha} = \Phi^T y$$

$\Phi \in \mathbb{R}^{N \times k}$

$\alpha \in \mathbb{R}^k$

$\hat{y}, y \in \mathbb{R}^N$

Solving the LS problem is equivalent to solving the system of linear eqⁿˢ

Normal eqⁿˢ : $\boxed{\Phi^T \Phi \lambda = \Phi^T y}$ $\quad \Phi^T\Phi \in \mathbb{R}^{k\times k}$

Solving a system of linear equations.

In general, Given $A \in \mathbb{R}^{k\times k}$, $b \in \mathbb{R}^k$ find $x \in \mathbb{R}^k$ s.t. $Ax = b$, Assume A is invertible.

$\boxed{x = A^{-1} b}$

Simple case: $U \in \mathbb{R}^{k \times k}$ s.t. U is upper-triangular.

Solve: $Ux = b$ for ($b \in \mathbb{R}^k$, U is invertible).

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$ ← upper triangular matrix

backward substitution

$Ux = b$, third eq^{n}: $u_{33} x_3 = b_3$

$$x_3 = \frac{b_3}{u_{33}}$$

$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$, $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ second eq^{n}: $u_{22} x_2 + u_{23} x_3 = b_2$

$$\Rightarrow x_2 = \frac{b_2 - u_{23} x_3}{u_{22}}$$

In general: $A \in \mathbb{R}^{k \times k}$, invertible. , $b \in \mathbb{R}^k$

Upper triangulars ← canonical

$A \xrightarrow{\text{transform}} U$

Elementary row operators

i) exchange of rows $i \leftrightarrow j$

ii) multiplication by a non-zero const. $(c \neq 0)$
cA_i, A_i : i^{th} row

iii) $A_i \leftarrow A_i + A_j$

$A \longrightarrow U$

Every elementary row operator corresponds to a matrix multiplication. by (elementary matrices)

A

exchange rows i & j

$$A = \begin{bmatrix} a_{11} & a_{12} \cdots & a_{1k} \\ a_{21} & a_{22} \cdots & a_{2k} \\ \vdots & & \\ a_{j1} & a_{j2} \cdots & a_{jk} \end{bmatrix} \longrightarrow \hat{A} = \begin{bmatrix} a_{j1} & a_{j2} \cdots & a_{jk} \\ a_{21} & a_{22} \cdots & a_{2k} \\ \vdots & & \\ a_{11} & a_{12} \cdots & a_{1k} \end{bmatrix}$$

$$E A = \hat{A}$$

$e_i^T = [0 \cdots \underset{\substack{\uparrow \\ i^{th} \text{ place}}}{1} \cdots 0]_{1 \times k}$

$e_i^T A = i^{th}$ row of A

$$E = \begin{bmatrix} - & e_k^T & - \\ - & e_2 & - \\ - & e_3 & - \\ & \vdots & \\ - & e_1^T & - \end{bmatrix}$$ ← exchange 1st & k^{th} row.

for a $c \neq 0$, $ce_k^T A$ ← k^{th} row of A by const c.

$$\begin{bmatrix} e_1^T \\ \vdots \\ e_{k-1}^T \\ ce_k^T \end{bmatrix} A$$

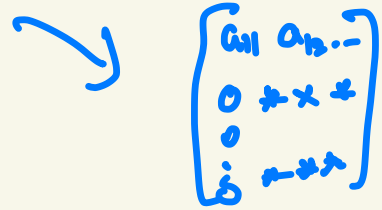
$(A_i \leftarrow A_i + A_j)$

$$\begin{bmatrix} e_1^T \\ \vdots \\ e_i^T + e_j^T \\ \vdots \\ e_k^T \end{bmatrix}$$ ← i^{th} row

$A \in \mathbb{R}^{k \times k}$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ a_{31} & a_{32} & \cdots & a_{3k} \\ \vdots & & & \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{bmatrix} \begin{matrix} \in R_1 \\ \in R_2 \\ \\ \\ \in R_k \end{matrix}$$

rows



Assume $a_{11} \neq 0$,

$$R_2 \leftarrow R_2 + \left(-\frac{a_{21}}{a_{11}}\right) R_1$$

$$R_i \leftarrow R_i - \frac{a_{i1}}{a_{11}} R_1 \qquad i = 2, 3, \cdots, k$$

$\cdots$ E_{31} E_{21} A

$$\begin{bmatrix} a_{11} & a_{12} & \cdots \\ 0 & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & & & \\ a_{m1} & \cdots & & a_{mk} \end{bmatrix}$$

2nd elementary row operation.

$l_{31} = -\frac{a_{31}}{a_{11}}$

1st elementary row.

$l_{21} = -\frac{a_{21}}{a_{11}}$

$$E_{k1} \cdots E_{31} E_{21} A = \begin{bmatrix} a_{11} & a_{12} \cdots & a_{1k} \\ 0 & * & * \\ \vdots & \vdots & \\ 0 & * \cdots & * \end{bmatrix}$$

$B_{(k-1)\times(k-1)}$

$$E_{k2} \cdots E_{42} E_{32} E_{k1} \cdots E_{21} E_{21} A = \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ 0 & & \\ \vdots & 0 & B_2 \\ 0 & & \end{bmatrix}$$

$(k-2) \times (k-2)$

step $(k-1)$ | step 2 | step 1

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ 0 & * & * \cdots * \\ \vdots & 0 & \ddots \\ 0 & 0 & 0 \quad * \end{bmatrix}$$

1 | $(k-2)$ | $(k-1)$

$$E_{21} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ l_{21} & 1 & \cdots & 0 \\ 0 & & & \\ \vdots & & & \\ 0 & \cdots & & 1 \end{bmatrix}_{k \times k}, \qquad E_{31} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ l_{31} & 0 & 1 & 0 & \cdots 0 \\ \vdots & & & & \\ 0 & \cdots & & & 1 \end{bmatrix}_{k \times k}$$

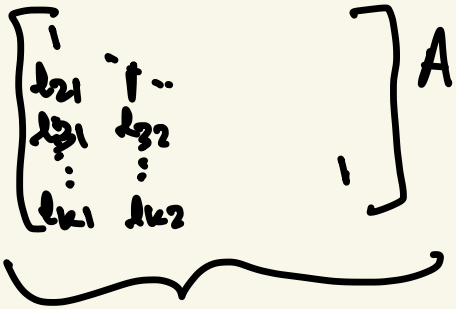
$$E_{21} = I + l_{21} e_2 e_1^T$$

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}$$

$$E_{31} E_{21}$$

$$= (I + l_{31} e_3 e_1^T)(I + l_{21} e_2 e_1^T)$$

$$= I + l_{31} e_3 e_1^T + l_{21} e_2 e_1^T = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ l_{21} & 1 & 0 & \cdots 0 \\ l_{31} & 0 & 1 & \cdots 0 \\ 0 & & & \ddots \\ \vdots & & 0 & \\ 0 & & & 1 \end{bmatrix}$$



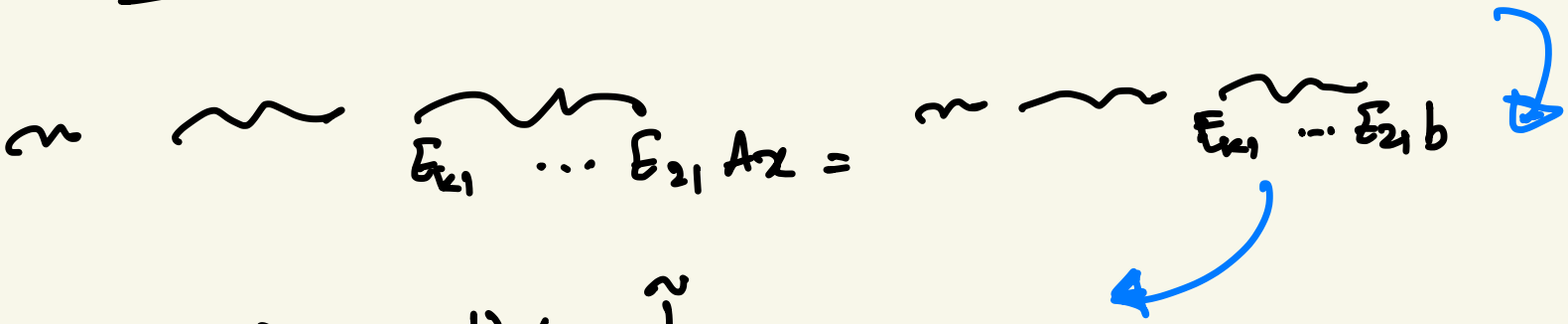
$A = U$

invertible-lower triangular

$$A = \begin{bmatrix} 1 & & \\ l_{21} & \ddots & 0 \\ \vdots & & \\ l_{k1} & & l_{k,k-1} \; 1 \end{bmatrix}^{-1} U$$

$\{ A = LU \}$ ← LU decomposition of A.

Problem: Solve $Ax = b$



$\Rightarrow \quad Ux = \tilde{b}$

A is invertible.

$A \in \mathbb{R}^{k \times k}$, then $A = LU$ where L is the lower traingular matrix with all 1's on the diagonal & U is the upper triangular matrix.

LU Decomposition of A

Uniqueness of LU decomposition.

$A \in \mathbb{R}^{k \times k}$ invertible. Then $A = LU$ such this decomposition is unique.

Pf: $A = L_1 U_1 = L_2 U_2$ with $L_1 = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ * & & 1 \end{bmatrix}$

$\Rightarrow L_1 U_1 = L_2 U_2$

$\Rightarrow L_2^{-1} L_1 = U_2 U_1^{-1}$

$L_2 = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ * & & 1 \end{bmatrix}$

$U_1 = \begin{bmatrix} \ddots & & * \\ & & \\ 0 & & \ddots \end{bmatrix}$, $U_2 = \begin{bmatrix} \ddots & & * \\ & & \\ 0 & & \ddots \end{bmatrix}$